

Fly High With SU2

The plane in the sky flies high and people looking at it marvel at its design. The complexities in designing the streamlined body of an aircraft are many. To make the designers' lives easier, we present here SU2 software that helps to simulate the design free of cost.

SU2—an open source suite

No one ever ventured into partial differential equation problems until it was realised how optimisation of problems through multi-physics strategy is possible via computational analysis. And that is where SU2 is introduced. It is penned down in the easiest platforms like C++ and Python. Being an open source freeware, it beats all the proprietary, unavailable and expensive software used over the years for designing aircraft systems.

Analyse, optimise and design

The streamlined body of an aircraft can be optimised using SU2 software. Each wing is optimised from upper to the lower end as per the requirement, while the right-hand side coloured column index on computer screen indicates pressure coefficients on each point. Thus, it carefully iterates the wing thickness and lift constraints in the aircraft as per calculations through differential equations.

Similarly, flow solution, adjoint solution, gradient calculations and mesh deformation

can be simultaneously carried out using SU2.

SU2 is conceptualised for supersonic aircrafts too. Not only simulation but also optimisation during aircraft design is monitored by SU2, where free-form deformation (FFD) control point variables keep a check on 3D area for shape deformation.

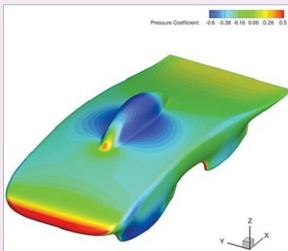


Fig. 1: Large-scale aircraft design using SU2 (Image courtesy: www.pointwise.com)

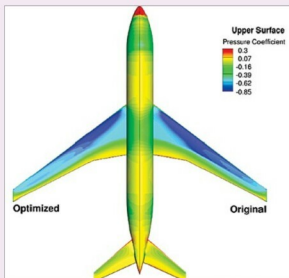


Fig. 2: Aircraft design—stream-wise pressure gradient (Image courtesy: www.pointwise.com)

SU2 5.0.0 The Raven

Within a year of upgrade, SU2 5.0.0 version has come up with special features that were not included in 4.3.0 cardinal.

SWIG with high-level API.

Now developers can wrap C++ functions for scripting languages.

So if you have some C++ functions, you can wrap them up in Python using SWIG accompanying high-level application programming interface (API).

Interpolation and transfer functions. Sampling and experimentation give innumerable data, which might be discrete. Introducing another independent variable into the data to estimate the value of a function is called interpolation. And this technique has been introduced for the first time in SU2, enabling designers to create impeccable designs.

Free form deformation technique. SU2 is a simulation software for solid geometric models using FFD technique. So if you have a deformed object and want to re-objectify from that hull object, you can do so using the FFD technique.

Harmonic balance flow and more feedback to user.

High-frequency oscillator circuits and complicated signal circuits can now be stabilised in SU2 software. For instance, in an amplifier, to gain steady state, the capacitor may be simulated hundreds to thousands of